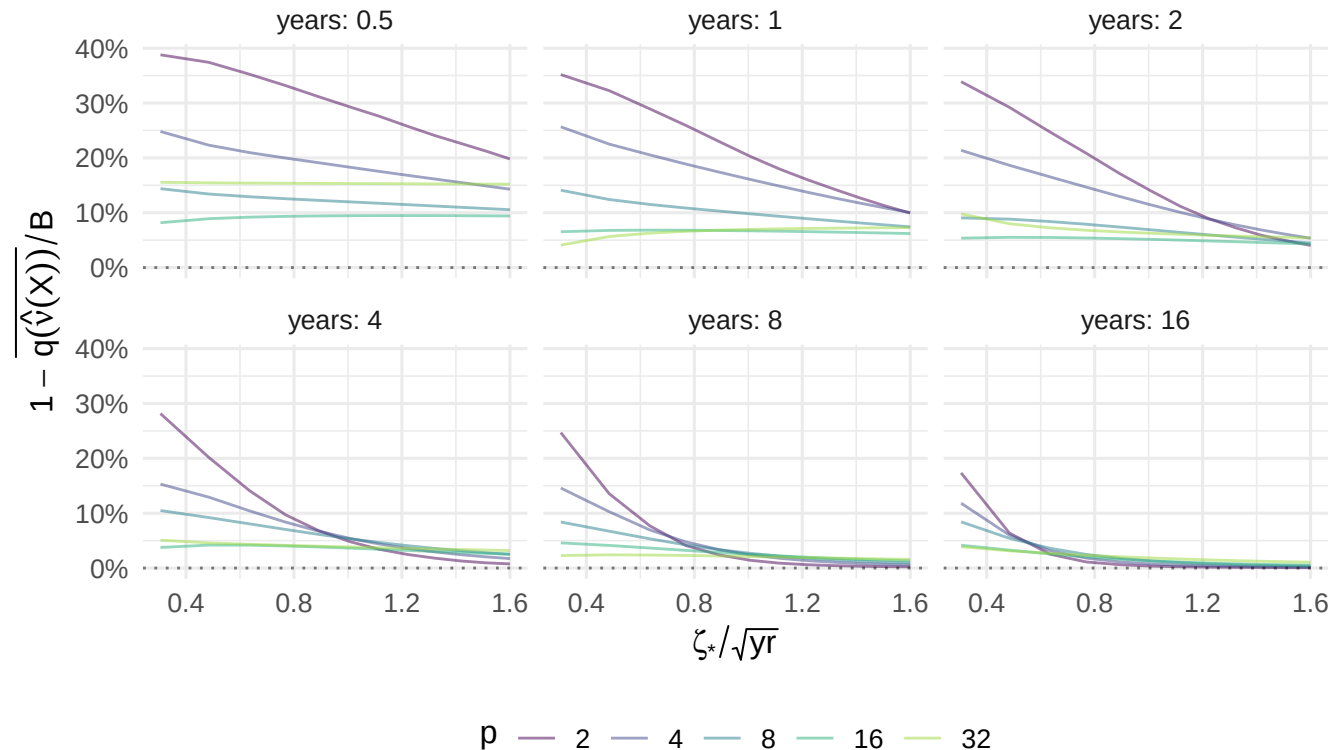


Relative Gap to Proposed Bound

Upper bound versus empirical expected signal noise of Markowitz portfolio



Using 5,000 simulations per configuration. Assumes 253 trading days per year.